

# P Pranav

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## Technical Skills

Languages: C++, Python, MATLAB

Robotics: ROS 2 (Humble/Jazzy), RViz, Gazebo, URDF, RTAB-Map, A\*/Dijkstra

Perception & Learning: OpenCV, YOLOv8, imitation learning (ACT), PyTorch, LeRobot

Tooling: Git, Docker, Linux

Hardware: Nvidia Jetson, Raspberry Pi, Arduino, ESP32, Intel RealSense

## Work Experience

**Indira Gandhi Centre for Atomic Research (IGCAR),  
Kalpakkam**

Jan 2026 - Apr 2026

*Bachelor Thesis Intern*

- Implemented a **visibility graph + A\* path planner for a 2-DOF SCARA manipulator**, running natively on the PLC motion controller and validated in real time on a 1:1 hardware test rig.
- Benchmarked A\* against Dijkstra across 1-10 obstacles resulting in identical optimal path lengths, with A\*'s compute-time advantage growing to 48% at 10 obstacles.
- Technologies used: Codesys, A\*, Dijkstra.

**Zoho Corporation, Chennai**

Sep 2024 - Feb 2025

*IoT Engineer Intern*

- Built an **ESP32 air-quality monitor** in embedded C++ reading 3 gas sensors, converting raw ADC readings to ppm through power-law calibration curves derived from the datasheets.
- Delivered a working prototype at ₹10k per unit against ₹20k-40k for commercial monitors, with OTA updates and alerts through Zoho Cliq and Mail.
- Technologies used: C++, ESP32, OTA updates, MQ-series gas sensors

**IGCAR, Kalpakkam**

Jun 2024 - Jul 2024

*Intern Trainee*

- Developed an **iterative Jacobian IK for a 2-DOF SCARA** in Python, adding speed-based time interpolation to cap joint velocities and remove the step changes in velocity that would trip the motors.
- Built an obstacle-avoidance planner with an interactive Pygame simulator, later extended into the multi-obstacle thesis planner.
- Technologies used: Python, NumPy, SciPy, Matplotlib, Pygame.

## Education

**Vellore Institute of Technology, Chennai**

2022 - 2026

**B.Tech. in Mechatronics and Automation**

CGPA: 8.61/10

Relevant Coursework: Python, C++, Machine Vision, Control Systems, Artificial Intelligence, Industrial Robotics.

## Academic Projects

**BT-101: Autonomous Manipulation Platform** ([github.com/ppranav04/BT-101](https://github.com/ppranav04/BT-101))

Independent, Ongoing

- Rebuilt joystick and leader-follower teleop for a **6-DOF SO-101 robot arm** pair from scratch and profiled loops at 293/145 Hz.
- Trained an **ACT policy** on 15 teleop demos with a single-camera setup and deployed it on the SO-101 arm.
- The policy reproduced approach-grasp-place but missed grasps; isolated 3 testable causes (monocular depth, start coverage, no recovery data).
- Stack: Python, PyTorch, LeRobot, scservo\_sdk, pygame, W&B, Hugging Face Hub

**Agriover: Vision-Guided Tomato Harvesting Arm**

VIT, Project-I

- Built the **perception-to-manipulation pipeline** in **ROS 2** (detection - IK - serial bridge - Arduino PWM) driving a 5-DOF 3D-printed servo arm.
- Localised fruit in 3D by deprojecting RealSense D435 depth through the camera intrinsics and a calibrated camera-to-base transform with 20.3 mm RMSE over a 0.15-0.50 m range.
- Fine-tuned YOLOv8n** on a custom tomato-ripeness dataset: F1: 0.91-0.92 on 784 test samples.
- Stack: ROS 2, Python, C++, YOLOv8, NVIDIA Jetson Orin Nano, RealSense D435